

Inherited retina functional testing and gene therapy

Part 2 | Match the recording to the clinical question

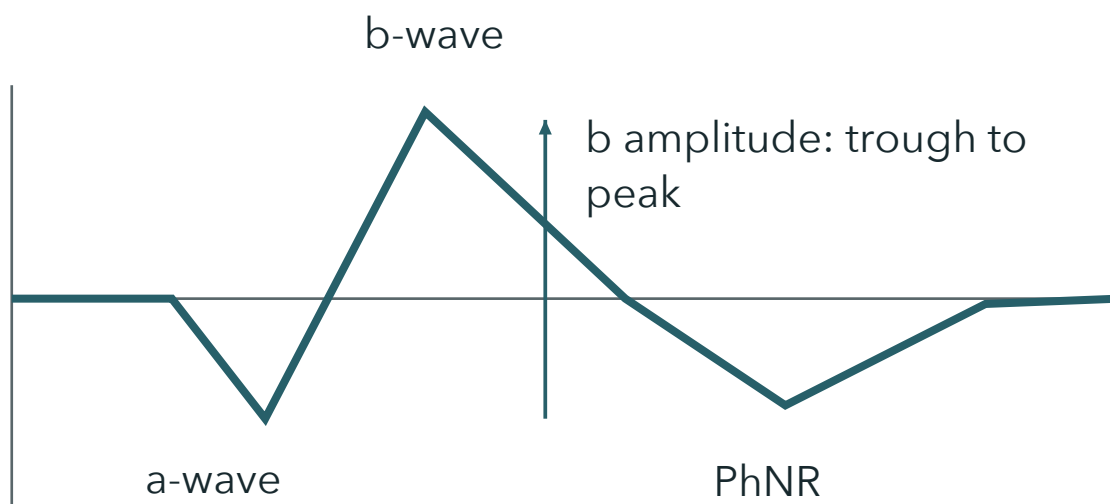
Normal global function does not exclude focal or interface disease.

Finding	Implication
Full-field ERG	Generalised rod and cone system function
EOG	RPE-dependent light rise at the photoreceptor interface
Pattern or multifocal ERG	Different spatial and cellular contributions
Structural imaging	Anatomy and viability rather than an electrical response

Start with the hypothesis, then choose the modality and confirm recording quality. A normal full-field trace cannot settle every macular or RPE disorder. State which function was sampled, what remains unsampled and whether adaptation, optics or electrodes could change interpretation. Complementary tests answer complementary questions rather than providing interchangeable retinal scores.

Read amplitude from the correct reference

Standard definitions make recordings comparable.



Original waveform: b-wave amplitude spans a-wave trough to b-wave peak.

A b peak of +105 microvolts and an a trough of -45 microvolts gives a b amplitude of 150 microvolts. Peak above baseline answers a different measurement question. Before localising dysfunction, confirm the stimulus, adaptation and reproducibility; otherwise an apparent biological difference may reflect protocol or recording conditions.

Best and Stargardt: interface and molecular distinctions

A macular phenotype should guide, not replace, functional and genetic assessment.

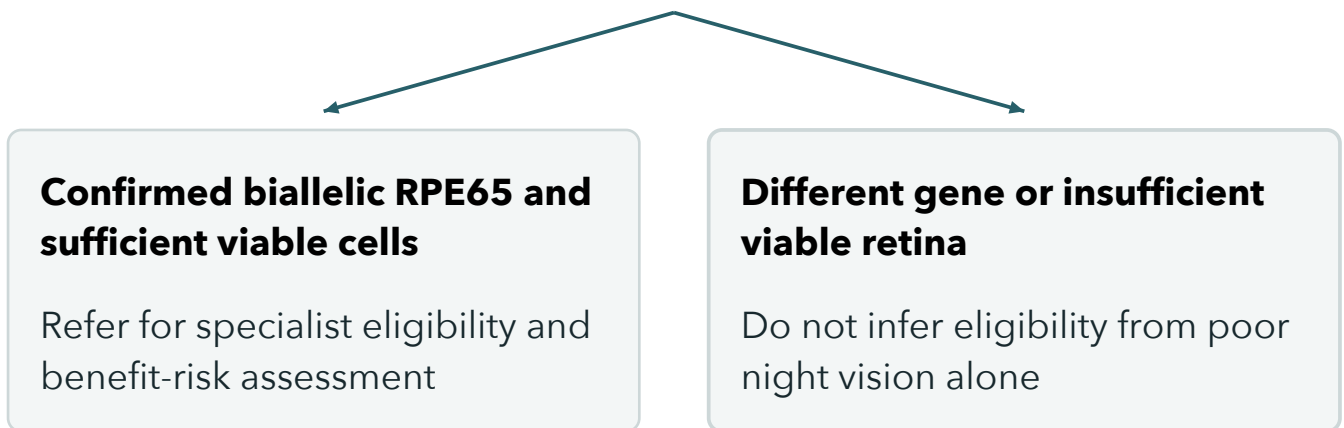
Finding	Implication
Bestrophinopathy	RPE-related dysfunction; EOG can be abnormal
Typical Best phenotype	Full-field ERG can remain preserved
Stargardt spectrum	ABCA4-associated disease has variable extent
Counselling	Confirm the molecular result and inheritance context

Describe the phenotype and the limits of each test before naming a mutation. Related appearances can have different genetic causes, and the extent of dysfunction can vary. The useful discussion connects OCT, autofluorescence, electrophysiology and molecular findings, then explains how uncertainty affects relatives, prognosis and potential trial eligibility.

RPE65 therapy: eligibility is not the phenotype label

Biallelic molecular confirmation and viable retina are required.

Does this inherited retinal disease meet the treatment indication?



Voretigene neparvovec is not a treatment for every inherited retinal degeneration. Confirm causative variants, viable retinal tissue and suitability for the delivery procedure. Explain the intended functional benefit and the difference between genetic correction in selected cells and reversal of advanced cell loss. A familiar syndrome name is not an adequate eligibility assessment.

Treatment discussion includes the procedure

A gene-directed product still has ocular delivery risks.

Finding	Implication
Intended benefit	Improved functional vision in eligible disease
Subretinal delivery	Specialist surgical and perioperative assessment
Damaged tissue	Limits recoverable function
Follow-up	Monitor response and ocular complications

Separate product eligibility from surgical suitability and patient priorities. Ask what change in real-world vision would matter, and explain that an anatomical or functional response does not imply normal sight. Discuss uncertainties, surveillance and rehabilitation alongside the intervention. Avoid presenting a molecular diagnosis as a promise of treatment success.